# Base station physics: summary

Params: base/spec/params.py v1.5.0

### Spring (each, A228, SF 1.2 on Shigley static limits)

| | |
|---|---|
| Rate used / by formula | 0.125 / 0.117 N/mm |
| Initial tension used | 7.0 N |
| Body coils, mass | 98, 53 g |
| Safe force (governed by) | 40.9 N (hook torsion) |
| Safe extension (wire) / spec cap | 271 / 265 mm |
| Surge (fixed-fixed) | 24 Hz |

### Launch (chain spring model)

| | |
|---|---|
| Stroke, ratio, pitch | 241 mm, 1.0:1, 35 deg |
| Spring extension full / rest | 265 / 24 mm |
| Stored energy (2 springs) | 12.1 J |
| Carriage force at full draw | 80 N |
| Muzzle speed | 6.87 m/s |
| Peak acceleration, stroke time | 18 g, 59 ms |
| Efficiency (dart KE / stored) | 55 % |
| Carriage KE at stop, pad peak | 1.90 J, 287 N (19 mm pad) |
| Spring KE at stop, coil clash | 0.83 J, 8.3 m/s |

### Flight (bead-chain tether + reel model, level ground)

| | |
|---|---|
| Line path at the shot | taut (path only lengthens) |
| Range at 0.25 / 0.5 / 0.75 N brake | 4.79 / 4.25 / 3.77 m (pitch 38) |
| Same shot, no tether | 5.16 m |
| Flight time, apex (0.5 N) | 0.91 s, 1.32 m |
| Impact speed, angle (0.5 N) | 6.2 m/s, 55 deg down |
| Ballistics table | 6 to 47 deg, best 38.5 deg: 4.25 m |

### Storage reel payout (LIGHT 0.25 N at full reel)

| | |
|---|---|
| Reel empty mass, inertia full | 98 g, 26.2 kg mm2 |
| Line radius full / core | 27.4 / 20.0 mm (brake +37 % at core) |
| Line pull in the stroke (spin-up) | 4.82 N |
| Reel rpm max | 2388 rpm |
| Overrun after landing | 1.37 m in 0.59 s |

### Capstan and tow

| | |
|---|---|
| Wraps for tow limit at the capstan (59 N, mu 0.15) | 2.6 raw, 4 design |
| Take-up needed for the tow at the actual wraps (7) | 0.08 N |
| Tow at limit: torque, speed, current | 3.27 N m, 0.142 m/s, 5.10 A |
| Wiper at limiter (5 N m), 12.6 V: torque, current, line pull | 5.0 N m, 7.0 A, 79 N (break 667 N) |
| Wiper motor stall (no limiter), 12.6 V: torque, current, line pull | 14.1 N m, 17.0 A, 223 N |
| Tow sensing at the dart: N per wiper amp above free | dart = 14.4 x (I - I_free) - 1.8 N (capstan side x1.18) |

### Drivetrain (wiper at 11.1 V with sag)

| | |
|---|---|
| aim: peak torque, current, time | 0.33 N m, 1.9 A, 7 s |
| cock: peak torque, current, time | 2.93 N m, 4.7 A, 3 s |
| reel: peak torque, current, time | 0.38 N m, 1.9 A, 106 s |
| tow: peak torque, current, time | 3.27 N m, 5.1 A, 35 s |

### To measure (then edit base/spec/params.py and run ./run_all.sh)

1. Spring hang test, each spring: hang 1, 2, 3 kg and measure the length each time (hooks included). python calib.py spring FREE_MM 1:MM 2:MM 3:MM, then SPRING k_n_per_mm and F0_n (mean of the two); the stronger spring decides the anchor holes (MEASURE_LIST B4). Every range number scales with the stored energy.
2. Wiper (MEASURE_LIST A6): free speed and current from a 40 % bench jog in telemetry, then WIPER n0_rpm = jog rpm x 2.5 (never the raw jog rpm) and free_a = amps. The stall cannot be read (jogs trip at 2.5 to 3 A, A7): keep stall_nm and stall_a.
3. Dart mass on a kitchen scale with ballast and tether knot (spec 0.279 kg), then DART mass_kg.
4. Payout brake (MEASURE_LIST G4, hanging weights: a luggage scale cannot read this low): in LAUNCH 20 g on the line must not pull line off and 30 g must (LIGHT 0.25 N); in NEUTRAL the FIRM clutch (5 N) lets line off at about 0.5 kg on the luggage scale.
5. First shots on level ground: python calib.py range pitch:brake:range ..., then LAUNCH v_scale = FACTOR.